

# I & Automation Enhancement

## AI Financial Risk Classification - Business Analysis Project

*Supplementary Enhancement Document | Version 1.0 | Prepared by Business Analysis Team*

**Purpose of this document:** This document supplements the existing BRD, FRD, User Stories, RACI, AS-IS/TO-BE process flows, and wireframes for the Financial Risk Classification solution. It does not repeat that content. It focuses solely on how AI-assisted decision support and process automation are proposed to enhance the existing solution.

## 1. AI & Automation Overview

The existing Financial Risk Classification solution, as defined in the approved BRD and FRD, evaluates loan applicants using structured business rules and historical customer data. This enhancement proposes layering AI-assisted decision support and process automation on top of that existing solution to further improve speed, consistency, and prioritization of loan officer effort - without changing the core business rules, scope, or governance already documented.

AI and automation are being proposed for this solution because the current rule-based process, while structured, still involves a high degree of manual effort in data checking, document review, and prioritizing which applications need closer attention. As application volumes grow, this manual effort becomes a bottleneck and introduces variability between reviewers. Introducing AI-assisted risk scoring and automated data handling is intended to address the following:

- Reduced manual effort - automating repetitive data validation and completeness checks currently performed manually by loan officers.
- Faster screening - enabling near-instant preliminary risk classification so loan officers can focus their time on applications that genuinely require judgment.
- Better risk prioritization - automatically surfacing and routing higher-risk applications to the right reviewer, rather than relying on manual triage.
- Improved consistency - applying the same AI-assisted logic and automated checks uniformly across all applications and branches, reducing reviewer-to-reviewer variability.

This document describes the proposed AI-assisted capabilities and automation opportunities at a business level. It is intended to sit alongside, not replace, the existing BRD and FRD.

## 2. Proposed AI Solution

The proposed AI-assisted solution is designed to support, not replace, the existing rules-based classification approach already defined in the FRD. It introduces an additional layer of AI-supported risk scoring intended to enhance the speed and depth of the initial risk read available to loan officers. The following capabilities are proposed:

### 2.1 AI-Assisted Risk Assessment

A proposed AI-assisted risk assessment capability that reviews applicant financial and demographic data alongside available historical data to generate a preliminary risk read, supplementing the existing rules-based classification with additional pattern-based insight.

## 2.2 Risk Classification: Low / Medium / High

The proposed AI-assisted solution would present applicants under the same Low / Medium / High risk classification structure already defined in the existing solution, ensuring consistency with current business rules, thresholds, and reporting categories rather than introducing a separate scoring scale.

## 2.3 Identification of Key Risk Factors

For each application, the proposed solution would surface the specific factors that most influenced the AI-supported risk read (for example, income-to-obligation ratio, repayment history, or employment stability), giving loan officers visibility into what is driving the classification rather than a single opaque score.

## 2.4 Explainable Risk Recommendation

The proposed solution emphasizes explainability: any AI-supported recommendation would be accompanied by a plain-language summary of the contributing factors, so loan officers and the Credit Committee can understand and evaluate the basis for the recommendation rather than treating it as a black box.

*Note: This is a proposed AI-assisted solution concept for business analysis and design purposes. No machine learning model has been built, trained, or deployed as part of this project, and no accuracy figures or production results currently exist.*

# 3. Process Automation

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In addition to AI-assisted risk scoring, the following process automation opportunities are proposed to reduce manual effort in the surrounding workflow:

- Automated data validation - system-driven checks of format, completeness, and consistency of applicant data as it is entered, reducing manual field-by-field review.
- Missing/invalid information detection - automatic identification of incomplete or inconsistent applicant data, with the application flagged back to the loan officer before it proceeds to classification.
- Risk-based application routing - automatic routing of applications to the appropriate reviewer or queue (loan officer approval, manual review, or Credit Committee) based on the assigned risk classification.
- High-risk application flagging - automatic flagging of high-risk applications for priority attention, ensuring they are not delayed behind lower-risk items in a reviewer's queue.
- Automated notifications - system-generated alerts to relevant users (loan officer, Credit Committee) when an application requires action or a decision has been recorded.
- Audit record creation - automatic creation of an audit trail entry for each classification, routing decision, and final outcome, supporting traceability and compliance review.

# 4. AI / Automated TO-BE Process

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The detailed AI / Automated TO-BE process flow for this enhancement has been developed separately in Lucidchart and is not reproduced here. Please refer to the linked/attached diagram.

## 5. AI & Automation Requirements

The following requirements define the business-level expectations for the AI-assisted and automated capabilities described in this document. These are supplementary to, and should be read alongside, the functional requirements already captured in the FRD.

| Requirement ID | Requirement                                                                                                                                                                             | Priority    |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| AIR-01         | The system shall automatically validate applicant data for completeness, format, and consistency before it proceeds to risk classification.                                             | Must Have   |
| AIR-02         | The system shall automatically detect and flag missing or invalid applicant information for correction by the loan officer.                                                             | Must Have   |
| AIR-03         | The system shall provide an AI-assisted risk assessment that supplements the existing rules-based classification with pattern-based insight from applicant and historical data.         | Should Have |
| AIR-04         | The system shall classify each applicant into the existing Low / Medium / High risk categories using the AI-assisted assessment alongside defined business rules.                       | Must Have   |
| AIR-05         | The system shall provide a plain-language explanation of the key factors contributing to each AI-supported risk recommendation.                                                         | Must Have   |
| AIR-06         | The system shall automatically route applications to the appropriate reviewer or queue based on the assigned risk classification.                                                       | Must Have   |
| AIR-07         | The system shall automatically flag high-risk applications for priority review.                                                                                                         | Must Have   |
| AIR-08         | The system shall require human review and final decision by an authorized loan officer or Credit Committee member for every application, regardless of the AI-supported recommendation. | Must Have   |
| AIR-09         | The system shall send automated notifications to relevant users when an application requires action or a decision has been recorded.                                                    | Should Have |
| AIR-10         | The system shall automatically create an audit record for each AI-supported classification, routing action, and final decision.                                                         | Must Have   |

## 6. Human-in-the-Loop

The proposed AI-assisted solution is designed around a human-in-the-loop model. AI is intended to support and accelerate the review process, not to replace human judgment or lending authority. Specifically:

- The AI-assisted capability provides a recommendation and risk classification (Low / Medium / High), along with the key factors behind it.
- A loan officer reviews the AI-supported recommendation, the underlying data, and the explanation provided before taking any action.
- The final lending decision - approve, decline, refer, or request further information - remains with an authorized human decision-maker (loan officer or Credit Committee), consistent with existing lending policy and the decision-authority rules already defined in the FRD.

This ensures that AI-assisted output is treated strictly as a decision-support input, with accountability for lending decisions remaining fully with authorized bank personnel.

## 7. Automation Opportunities

| Process                         | Current Activity                                                                           | Proposed Automation                                                                              | Benefit                                                              |
|---------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Applicant data intake           | Loan officer manually reviews entered data for completeness and correctness.               | Automated, real-time data validation as data is entered.                                         | Fewer errors reaching classification; less manual rework.            |
| Incomplete application handling | Loan officer identifies missing information through manual review, often after submission. | System automatically detects and flags missing/invalid fields at point of entry.                 | Faster correction; fewer delayed or returned applications.           |
| Initial risk screening          | Loan officer manually assesses financial data alongside rule-based classification output.  | AI-assisted risk read generated automatically alongside the existing rules-based classification. | Faster initial screening; more consistent starting point for review. |
| Application prioritization      | Applications reviewed largely in order received, regardless of risk level.                 | Risk-based routing and automatic high-risk flagging to prioritize queues.                        | Higher-risk applications receive attention sooner.                   |
| Reviewer/committee alerts       | Loan officers and committee members check status manually or wait for manual follow-up.    | Automated notifications triggered by status changes and required actions.                        | Reduced turnaround time; fewer missed follow-ups.                    |
| Audit documentation             | Audit trail information compiled manually or reconstructed after the fact.                 | Automatic audit record creation for every classification, routing action, and decision.          | Stronger, always-on auditability with less administrative effort.    |

## 8. Expected Business Benefits

- Reduced manual effort - less time spent on manual data checking and completeness review.
- Faster application screening - quicker initial risk read available to loan officers at the point of submission.
- Improved consistency - uniform application of AI-assisted classification and automated checks across all branches and reviewers.
- Better risk prioritization - high-risk applications are surfaced and routed sooner rather than waiting in a standard queue.
- Improved transparency - explainable, factor-level rationale accompanying each AI-supported recommendation.
- Better auditability - automatically generated, complete audit records for every classification and decision.

*These are expected, proposed benefits of the enhancement concept. As no solution has yet been built or deployed, they have not been measured or confirmed against actual results.*

## 9. Proposed KPIs

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The following KPIs are proposed to measure the effectiveness of the AI-assisted and automation enhancements once implemented. Baseline and target values would be established during a future measurement phase and are not defined here.

- Application processing time - end-to-end time from submission to final decision.
- Manual review time - average time spent by loan officers/Credit Committee reviewing an individual application.
- Automated validation rate - proportion of applications passing data validation without manual intervention.
- High-risk identification rate - proportion of high-risk applications correctly flagged and routed for priority review.
- Exception rate - proportion of applications requiring manual override or exception handling outside standard automated flow.
- AI recommendation override rate - proportion of AI-supported recommendations changed by a human reviewer during final decision-making.